

Professional Self-Assessment

The Computer Science (CS) degree surpassed my initial expectations when starting this program. I have personal prior experience and professional certificates in Web Design. My goal when starting this program was to continue in that direction, creating websites for side projects and hoping one becomes successful. I want to continue work on my personal passion projects, but in the meantime, the CS program has changed my professional life.

Shortly after starting my computer science studies there was an opportunity in my company to help one of the software engineering teams part-time. I jumped on this chance. What started with a few hours a week, became half-time, then full-time, then a job offer to become a software engineer. At this point, about halfway through my CS program, I felt I had a solid grasp on programming. As the saying goes, “You don’t know what you don’t know.” Through my university coursework we experienced multiple programming languages, whereas at my work we develop primarily in Java. We use Git at my work, but we do not have strict processes for version control. In a class on collaborating, we learned the formal version control features of Git such as branching for development, production, and troubleshooting, and using pull requests. At my work we use Git in conjunction with Artifactory, Jenkins, and Maven for a Continuous Integration/Continuous Development (CI/CD) pipeline for development. However, a senior developer always managed our build processes. After university coursework I was able to step in when the senior developer was unavailable, creating a new build for an internal customer. I used MySQL in my personal web design work, but I did not understand databases. Now through coursework I have working knowledge of databases including MySQL, PostgreSQL, MongoDB, and DynamoDB.

As I mentioned previously, my personal goals for this Computer Science program are to learn the skills and get experience with creating full-stack web applications. I have multiple ideas for web applications that I would like to develop. Thus, I took multiple courses on full-stack web applications, mobile applications, and User Interface/User Experience (UI/UX). These courses were for my personal goals, but I found they benefited my professional work as well. I was working on system integration testing on a major product. There was an intermittent bug that was causing errors between the Graphic User Interface (GUI) and the back-end controller. We have separate teams that develop the GUI and the back-end controllers. Both teams tested their systems independently and could not isolate the error. I was able to use my experience from testing the back-end along with some Angular skills I learned in my full-stack courses to troubleshoot the issue to a race condition between the systems. Working with the GUI and back-end team, we fixed the bug

and delivered a new build to the customer. These stories demonstrate one of the most important lessons I have learned; you can never know all there is to know because there will always be new challenges in Computer Science.

This ePortfolio consists of a complete full-stack web application I built for an idea I have had for a few years. The web application allows users to search for new build home communities. I wanted to make a simple website that would not collect user data or be plagued with advertisements. I had this idea when I was looking for a new home and I could not find a single source online that had all the information from multiple builders. I was excited to use this opportunity to bring this idea to life. I felt this project would be a perfect environment to showcase the development skills that demonstrate the course outcomes. The initial artifact I chose for this course was an initial draft of the full-stack web application I wanted to build, but this initial draft was created completely by generative AI. It was a good starting point but had many errors and design issues. The artifact consists of many files that make a full-stack web application including a React front-end, back-end served by Express, with a PostgreSQL DB handled by Prisma. Each enhancement included multiple files in this stack and touched on all the course outcomes to some degree.

First Enhancement

I added a new interface so that home builders could log into the web application and add new communities to the database. Prior to this enhancement the application had no concept of user roles. I added an admin role (me) that could approve other home builder accounts. Home builders could then register, and after the admin approves them, add their own communities to the database through a modal interface. This enhancement demonstrated the following course outcomes:

Collaborating in a team environment – added comments throughout to make the code easier to understand and maintain. Admin accounts make it easier to add other team members as the project grows.

Communicating with stakeholders – added the customer interface to make it easier for customers to interact with the website.

Software engineering and database – to make the customer interface I had to add multiple new modals to the front-end, routes to the back-end, and users to the database.

Security – user accounts use Json Web Tokens (JWT) for authentication. The admin account uses credentials that are hashed for security. Only admin approved user accounts can add new communities. The form to add new communities validates all inputs.

Second Enhancement

I added a new search by location filter and sort. This required adding a location option to the search filter. This addition was straightforward but would only return communities that matched that location. Ultimately, the application needed to find communities near that location. Next, the sort feature needed to be able to sort by how close communities are to the desired location. This feature required adding a new algorithm that could do the calculations for distance and then compare distances. This algorithm is in the newly created haversine function.

Collaborating in a team environment – added comments throughout to make the code easier to understand and maintain.

Communicating with stakeholders – searching and sorting by location is a critical feature for customers.

Data Structures and algorithms – the core of this enhancement is the new algorithm to calculate the distances and sort by distance from the desired location.

Software engineering and database – required updating the filtering and sorting logic.

Security – validates the filtering inputs including the location.

Third Enhancement

I updated the database schema to remove the Home model and incorporate all the information into the Community model. I also seeded the database with 60 example communities for testing. The primary change was in the database files, but there were ripples throughout the full-stack to accommodate the database changes.

Collaborating in a team environment – added comments throughout to make the code easier to understand and maintain. Seeding the database makes it easier for teams to test other functions of the application. Combining models into one makes it easier to maintain the database as it grows.

Communicating with stakeholders – other home websites are bloated with information. By removing the Home model and adding simple, filterable ranges to the Community model, the overall experience is simplified for customers.

Data Structures and algorithms – adding 60 communities to the database enabled testing of the many filter and sort options and their underlying data structures and algorithms.

Software engineering and database – removed the Home model and added the necessary information to the Community model, simplifying the database structure. Had to make many updates throughout the application to accommodate the database change.